

Supplemental Material for
“Role differentiation as ignition of a collective information
engine”

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NOTATION

Symbols are grouped so that quantities separated by a single averaging step, or by one stage of the loop, sit together. Equation numbers without an S prefix refer to the main text.

Status and its differences. Distinguished by which partner information has been averaged away.

x_i	Status of agent i in context m , the projection $\boldsymbol{\omega}_i \cdot \mathbf{g}_m$.
$\Delta_{im}(j)$	Difference to a named partner, $(\boldsymbol{\omega}_i - \boldsymbol{\omega}_j) \cdot \mathbf{g}_m$; what one encounter presents.
Δ_{im}	After the partner average, $(1 - \rho_{\text{pair}})x_i$; what enters Eq. (2).
Δ	Pairwise difference at fixed context and partner (Appendix A, Sec. S8).
s_{im}	Observed signal, $\Delta_{im}(j) + \sigma_{\text{read}}\xi$; the only status an agent sees.

Action and policy. Distinguished by which noise has been averaged away.

a	Action, -1 (Defer) or $+1$ (Go).
T	Evidence samples per decision; the read memory.
π	Action policy, $p(a \mid \{s_{im,t}\}_{t=1}^T)$.
$\bar{\pi}(\Delta)$	Go-probability at true difference Δ ; the policy averaged over read noise.
$\bar{a}(\Delta)$	Noise-averaged action at a <i>given</i> partner, $2\bar{\pi}(\Delta) - 1$; odd in Δ .
$\langle a \rangle_m(\boldsymbol{\omega}_i)$	Noise- and partner-averaged action; the response in Eq. (2).

Gains. One per stage, one for the closed loop.

g	Read gain, $2\pi'(0) = \sqrt{2T/\pi}/\sigma_{\text{read}}$; slope of $\bar{a}$ at zero difference.
η_{agent}	Inference efficiency, $\sigma_{\text{read}}g/2$; equals $\sqrt{T/2\pi}$ for T samples.
$\mu_{\text{eff}}(v)$	Read gain softened by partner variance v ; $\rightarrow g$ as $v \rightarrow 0$.
gu_m	Axis m 's share of the loop gain: read times write, each $\propto \sqrt{u_m}$.
$\mathbf{\Lambda}$	Social loop gain matrix, Eq. (5).
Λ_k	Its k -th eigenvalue; axis k ignites at $\Lambda_k = 1$.

Noise. Two independent sources, and what the reduction discards.

σ_{read}	Observation (read) noise; the finest status difference resolved.
σ_{ω}	ω -identity noise; variation in $\boldsymbol{\omega}_i$ from outside the modeled encounters.
$\xi, \boldsymbol{\eta}_i$	Standard Gaussian sources for the two noises above.
v	Partner-projection variance $\mathbf{C}_{mm}$, of order σ_{ω}^2 subcritically.
$\beta b_0 = \beta^2/\nu$	Shot-noise intensity discarded by the dense-encounter limit.

State and its spectra.

ω_i	ω -identity of agent i , a vector in $\mathbb{R}^d$.
Ω	The N_a ω -identities stacked.
r_i	Accumulated coordination income (wealth) of agent i .
$\mathbf{g}_m, \hat{\mathbf{g}}_m$	Schema vector and its direction, $\mathbf{g}_m = \sqrt{u_m} \hat{\mathbf{g}}_m$.
u_m	Schema strength $ \mathbf{g}_m ^2$; the power axis m puts on the channel.
$\mathbf{H}$	The $M \times d$ matrix of stacked schema vectors.
$\mathbf{C}$	Population covariance $N_a^{-1} \sum_i \omega_i \omega_i^\top$; the order parameter.
$\mathbf{C}_r$	Resource-weighted covariance, $\sum_i r_i \omega_i \omega_i^\top / \sum_i r_i$.
v_k	k -th eigenvalue of $\mathbf{C}$; leaves the floor $\sigma_\omega^2/2\gamma$ when k ignites.
v_m	Variance of $\mathbf{C}$ along $\hat{\mathbf{g}}_m$, $\hat{\mathbf{g}}_m^\top \mathbf{C} \hat{\mathbf{g}}_m$; equals v_k for orthogonal contexts.
μ_k	k -th eigenvalue of $\mathbf{H}^\top \mathbf{H}$; equals u_k for orthogonal contexts.
λ_m	Resource-weighted variance along $\hat{\mathbf{g}}_m$, $\hat{\mathbf{g}}_m^\top \mathbf{C}_r \hat{\mathbf{g}}_m$.

Rates.

γ	ω -identity decay rate; $1/\gamma$ is the write memory.
ν	Encounter rate per agent, a fixed budget independent of N_a .
b_0	Per-encounter kick to the ω -identity.
$\beta = \nu b_0$	Feedback strength; the rate decisions are written into ω -identity.
c	Resource decay rate; $1/c$ the income horizon.
α, κ	Schema reinforcement and decay rates.
$\varphi(U)$	Budget pressure of the shared channel, common to every schema.
$r_{\text{comp}}, r_{\text{act}}$	Compounding and activation rates of the u -transient; $r = r_{\text{act}}/r_{\text{comp}}$.
τ_k	Relaxation time of mode k , $1/\gamma 1 - \Lambda_k $.

Population and pairing. Three different counts of agents.

N_a	Number of agents.
N_{enc}	Encounters in one coarse-graining window, $\nu \Delta t$ (Sec. S8).
N_{eff}	Participation ratio of the pairing kernel; $N_a - 1$ if uniform.
f_m	Play frequency of context m , with $\sum_m f_m = 1$.
$p_{ij}^{(m)}, \mathbf{P}^{(m)}$	Pairing probabilities set by the matching rule, and their kernel.
ρ_{pair}	Correlation of a partner's status with one's own, set by that rule.
$\mathbf{L}^{(m)}$	Graph Laplacian $\mathbf{I} - \mathbf{P}^{(m)}$; how a persistent network would enter.

Repertoire and capacity. Four counts of schemas, by what limits each.

M, d	Number of contexts and embedding dimension.
$q = M/d$	Aspect ratio of the repertoire.
R	$\text{rank}(\mathbf{H}) \leq \min(M, d)$.
u_0, σ_u	Seed scale and log-dispersion, $u_m(0) = u_0 e^{\sigma_u z_m}$.
u_{cap}	Per-axis range ceiling of the channel.
$U = \sum_m u_m$	Total channel power, limited by the capacity U_{cap} .
M_{cap}	$U_{\text{cap}}/u_{\text{cap}}$; axes drivable to full range at once.
Θ	Capacity ratio $(U_{\text{cap}}/M)/u_{\text{cap}} = M_{\text{cap}}/M$; share per schema in ceiling units.
M_{eff}	Axes clearing the context decay rate at a self-consistent fixed point, $\lambda_k(\mathbf{C}_r) \geq \kappa/\alpha$.
K	Realized survivor count at the end of the u -transient; need not saturate.
$N(\beta)$	Cultural thickness; active role axes at feedback strength β .

Payoff and information.

w_0	Maximal coordination gain per encounter.
W_{ijm}, W_{im}	Pairwise payoff above Nash and its partner average.
$P_{e,ijm}, P_{e,im}$	Pairwise and partner-averaged role-inference error rates.
I	Role information $1 - H_{\text{bin}}(P_e)$, which bounds the payoff.
$\bar{r}_{\text{rf}}$	Role-free resource mean, of order σ_ω^2 .

Terms. Defined terms without a symbol of their own; those with one are glossed in the tables above.

Structuration.

<i>structuration</i>	Giddens's resolution of that debate: structure as medium and outcome.
<i>duality of structure</i>	Structure is both the medium and the outcome of what it organizes.
<i>structure–agency debate</i>	Whether structure shapes individuals or their choices generate it.
<i>schemas</i>	Transposable rules and conventions that prescribe behavior around resources.
<i>resources</i>	Material and symbolic stocks that schemas mobilize and that empower users.
<i>transposability</i>	That a schema carries across contexts; here the overlap $\hat{\mathbf{g}}_m \cdot \hat{\mathbf{g}}_n$.
<i>institutionalization</i>	A schema gaining strength; allocation of channel capacity across games.

Engine and channel.

<i>status channel</i>	The collective self-sensing process by which agents read status differences.
<i>status</i>	Label for the signal; its reward asymmetry is immaterial to the theory.
<i>ignition</i>	Onset of the instability at $\omega = 0$; roles then sustain themselves.
<i>irreducible cognitive floor</i>	$\eta_{\text{agent}} > \sigma_{\text{read}}\gamma/4\beta$; below it no matching rule sustains roles.
<i>well-mixed</i>	Uniform pairing, $p_{ij}^{(m)} = 1/N_{\text{a}}$; the Laplacian becomes a projector.
<i>platform-owned</i>	Set by the operator rather than by the agents; all factors but η_{agent} .

Repertoire dynamics.

<i>strength sector</i>	The u_m half of the polar split of the schema dynamics; solved here.
<i>transformation sector</i>	The $\hat{\mathbf{g}}_m$ half, rotating what a schema is about; left exogenous.
<i>u-transient</i>	Institutionalizes the repertoire: winnowing, then shaping.
<i>active set</i>	$\{m : u_m > u_0\}$, the schemas above the seed scale; its size is K .
<i>invasion threshold</i>	$\alpha\lambda_m = \kappa + \varphi$, above which a schema's strength grows.
<i>thick band</i>	$\Theta \geq 1$, where the per-schema ceiling binds first and all M survive.
<i>saturated cluster</i>	Endpoint with the survivors together at the ceiling u_{cap} .
<i>condensation</i>	Runaway of all strength onto one schema when the budget cannot hold it.
<i>pinning</i>	Arrest of that runaway at the ceiling, with a fringe at the threshold.

Cascade classes. The five spectral shapes of $\mathbf{H}^\top \mathbf{H}$.

<i>degenerate</i>	Equal eigenvalues; one avalanche, with no advance warning.
<i>geometric</i>	Constant ratios; a logarithmic cascade, the most monitorable class.
<i>Zipf</i>	Power-law strengths; decelerating for $a > 1$, accelerating for $a < 1$.
<i>spiked</i>	One outlier over a bulk, from coherent contexts sharing directions.
<i>random</i>	Marčenko–Pastur bulk; quiet, then a burst at the upper edge.

Matching rules and memory modes.

<i>random pairing</i>	Partner drawn independently of status, $\rho_{\text{pair}} = 0$; the reference case.
<i>rank-ordered pairing</i>	Each rank paired with its mirror, $\rho_{\text{pair}} \rightarrow -1$; halves onset.
<i>assortative pairing</i>	Like paired with like, $\rho_{\text{pair}} \rightarrow 1$; the gain factor vanishes.
<i>feature mode</i>	Graded projections near the bifurcation, soft and high-dimensional.
<i>prototype mode</i>	Sharp, near-saturated projections deep in the role phase.
<i>bimodal split</i>	The population occupying complementary basins rather than one.

S1. THE MODEL LADDER AND THE COORDINATION PAYOFF

A. The ladder of six models

The main text presents the closed loop directly. It was developed as a sequence of six models of increasing structure, each adding one ingredient and exposing one phenomenon. We summarize them here (Table S1). The reductions identify which ingredient produces which result.

TABLE S1. The six models. Number of agents (N_a) and number of contexts (M). Each adds one ingredient (bold) and yields one key result. Model 1 has exogenous feedback; the rest are endogenous.

# N_a / M	New ingredient	Key result
1 1 / 1, ex.	policy π , external oracle	classical info engine; $\bar{\pi}'(0)$ is measurement quality
2 2 / 1,	reciprocal measurement	self-generated fuel; $G = 2g$, $\beta_c = \gamma/2g$
3 $N_a / 1$	population, mean field	phase transition; $G = g$; order parameter C
4 N_a / M	vector roles	$\Lambda = \Lambda \mathbf{H}^\top \mathbf{H}$; conditional cascade
5 N_a / M	dynamic schemas	self-consistency; but schemas collapse to one axis
6 N_a / M	resources	replicator on fixed axes; endogenous cascade

Model 1 places a single agent in a Go/Defer game against a fixed environment. An oracle supplies a noisy role signal $s = \Delta + \sigma_{\text{read}}\xi$, with ξ a standard Gaussian noise source. The agent's action policy $\pi : \mathbb{R} \rightarrow [0, 1]$ maps signal to Go-probability. A policy that acts on the sign of a single observation has slope $\bar{\pi}'(0) = 1/(\sigma_{\text{read}}\sqrt{2\pi})$; pooling T signals before thresholding gives $\bar{\pi}'(0) = \sqrt{T}/(\sigma_{\text{read}}\sqrt{2\pi})$. The model is a Szilard engine, with the oracle as the measurement, the policy as the feedback protocol, and the decision as work extraction. The fuel is $I(\text{sign}(\Delta); s) = 1 - H_{\text{bin}}(P_e)$ bits and the work is $W = w_0(1 - 2P_e)^2$, derived in Sec. S1 B. The two satisfy $W \leq 2 \ln 2 w_0 I$ (equivalently $W \leq 2w_0 I$ with I in nats), so a coordination payoff requires role information. The bound is saturated as $P_e \rightarrow \frac{1}{2}$, and both W and I vanish in the symmetric (undifferentiated) state. Ref. [1] gives the corresponding bound on consensus (flocking) order in informational active matter. Both follow from the feedback second law of Sagawa and Ueda [2].

Model 2 removes the oracle: two agents with statuses x_1, x_2 each measure the other, so the fuel is self-generated. The status difference $\Delta = x_1 - x_2$ obeys $\dot{\Delta} \approx (-\gamma + 2\beta g)\Delta + \sqrt{2}\sigma_\omega\eta$, with gain $G = 2g$ (a factor 2 from the $\pm\beta$ swing, another from reciprocal amplification) and threshold $\beta_c = \gamma/2g$. For $\beta < \beta_c$ the difference decays to zero; for $\beta > \beta_c$ it grows until the cubic term saturates it.

Model 3 takes N_a agents randomly paired, in the mean field. Only the focal agent's update contributes, halving the gain to $G_{\text{MF}} = g$, so $\beta_c^{\text{MF}} = \gamma/g = \gamma\sigma_{\text{read}}\sqrt{2\pi}/2$. The mean $\langle x \rangle$ vanishes in the anti-coordination phase, so the order parameter is the population variance $C = N_a^{-1} \sum_i x_i^2$, with $C - C_0 \propto (\beta - \beta_c)$ near threshold.

Model 4 gives agents vector positions $\boldsymbol{\omega}_i \in \mathbb{R}^d$ and M contexts $\mathbf{g}_m \in \mathbb{R}^d$. The loop gain becomes a matrix on the context second moment, $\boldsymbol{\Lambda} = \Lambda \sum_m \mathbf{g}_m \mathbf{g}_m^\top = \Lambda \mathbf{H}^\top \mathbf{H}$, with $\mathbf{H}$ the $M \times d$ matrix of stacked context vectors (Sec. II A of the main text). Its eigenvalues give the cascade $\beta_c^{(k)} = \gamma/\mu_k$. The order parameter generalizes to the covariance $\mathbf{C} = N_a^{-1} \sum_i \boldsymbol{\omega}_i \boldsymbol{\omega}_i^\top$, with the number of supra-floor eigenvalues—the cascade height $N(\beta)$ —the cultural thickness.

Model 5 lets the context directions evolve, $\dot{\mathbf{g}}_m = -\kappa \mathbf{g}_m + \alpha \mathbf{C} \mathbf{g}_m$: a power iteration rotating $\mathbf{g}_m$ toward the leading eigenvector of $\mathbf{C}$. Schemas align with the role structure and the role structure reinforces the schemas, which is one half of Sewell's duality. All schemas converge on the same axis, so the stationary structure is one-dimensional. The same collapse occurs for the resource-weighted rotation of Model 6, and is analyzed with its two candidate remedies in Appendix D of the main text.

Model 6 closes the loop with resources (Sec. II B of the main text): schemas are reinforced by the resource-weighted covariance $\mathbf{C}_r$, so agents who profit from a schema defend it. Reinforcement is thereby contingent on success rather than on raw differentiation, and schema competition becomes a replicator dynamics whose fitness is resource accumulation. The context directions are held fixed (Sec. I of the main text), so what the loop selects is the strength of each schema, not its axis.

B. The coordination payoff

The oracle (or, endogenously, the population) assigns a leader (should Go) and follower (should Defer). Each agent's signal has independent role-inference error rate P_e . The four

joint outcomes and their payoffs (averaged over the pair) are: both correct, prob $(1 - P_e)^2$, payoffs (6, 2), average 4; leader wrong only, prob $P_e(1 - P_e)$, both Defer (3, 3), average 3; follower wrong only, prob $(1 - P_e)P_e$, both Go (-1, -1), average -1; both wrong, prob P_e^2 , roles swap to (2, 6), average 4—still coordinated. The expected average payoff is

$$\bar{U} = 4(1 - P_e)^2 + 3P_e(1 - P_e) - (1 - P_e)P_e + 4P_e^2 = 4 - 6P_e(1 - P_e). \quad (\text{S1})$$

The symmetric mixed equilibrium has each player Go with probability p^* fixed by indifference, $-p^* + 6(1 - p^*) = 2p^* + 3(1 - p^*)$, so $p^* = \frac{1}{2}$ and both actions pay $U_{\text{Nash}} = \frac{5}{2}$. Subtracting it and using $6P_e(1 - P_e) = \frac{3}{2} - \frac{3}{2}(1 - 2P_e)^2$,

$$W = \bar{U} - U_{\text{Nash}} = w_0 (1 - 2P_e)^2, \quad w_0 = \text{CE} - \text{Nash} = 1.5. \quad (\text{S2})$$

The quadratic form is specific to two-role anti-coordination: a double error swaps the roles and leaves the pair coordinated (the fourth outcome has the same payoff as the first), so the work depends only on $(1 - 2P_e)^2$. Because W is quadratic in P_e , the resource income of Sec. II B of the main text is quadratic in ω -identity, which allows the Gaussian fixed-point calculation to close.

C. Role-following as a self-enforcing correlated equilibrium

The role signal acts as a correlating device [3, 4]. At each encounter the noisy status $s = (\boldsymbol{\omega}_i - \boldsymbol{\omega}_j) \cdot \mathbf{g}_m + \sigma_{\text{read}}\xi$ recommends the action $a_i = \text{sign}(s)$ to agent i . The obedience constraint is that following this recommendation is a best response when the partner also follows. Writing $q = \text{Pr}(\text{partner plays Go} \mid s_i)$, the anti-coordination payoffs give expected returns

$$\text{Go: } 6(1 - q) - q = 6 - 7q, \quad \text{Defer: } 2q + 3(1 - q) = 3 - q, \quad (\text{S3})$$

so a Go recommendation is obedient for $6 - 7q \geq 3 - q$, that is $q \leq \frac{1}{2}$, and a Defer recommendation for $q \geq \frac{1}{2}$. The status difference is antisymmetric: an agent recommended Go is the higher-status party, whose partner is the more likely to Defer. Both constraints therefore reduce to the same requirement, that the recommendation anti-correlate with the partner's action. This is a condition on measurement quality, set by $\Delta/\sigma_{\text{read}}$ and the integration T , the same quantities that enter η_{agent} .

The equilibrium is not in dominant strategies. Following is a best response only if the partner also follows: against a partner who ignores the signal and always plays Go, the best response is to always Defer. Role-following therefore requires shared adoption. That adoption need not be assumed: replicator dynamics on strategies, that is imitation of higher-paying play, selects the correlated equilibrium in social dilemmas [5], and the same selection would make following an outcome here. Because the correlating device is noisy, the applicable solution concept is correlated equilibrium of the Bayesian game [6]. The policy is not a further assumption. The payoff depends on it only through the error rate, $W = w_0(1 - 2P_e)^2$, and P_e is smallest when an agent reports the sign of its accumulated evidence. Lowering P_e raises $\bar{\pi}'(0)$, and with it η_{agent} . Against a partner who also follows, the policy used here is therefore the one any algorithm that maximizes reward would find. This holds where the cost of the memory supporting the inference is negligible against the payoff it yields. Once that cost is counted, the strategy that minimizes P_e need not maximize net output [7].

The obedience constraint tightens as roles differentiate. At the symmetric fixed point ($\Delta \rightarrow 0$) the signal carries no information, $q = \frac{1}{2}$, and both constraints hold with equality. The correlated equilibrium then coincides with the mixed Nash equilibrium and pays 2.5. As Δ grows the signal becomes informative, $q < \frac{1}{2}$, and the role-signal equilibrium pays 4 and is strictly self-enforcing. Ignition ($\Lambda > 1$) is the instability that takes the population from the first case to the second: the order parameter and the obedience slack $\frac{1}{2} - q$ grow together, both set by the informativeness of the signal. The correlating device here is the agents' own accumulated status rather than an external mediator.

The calculation is carried out at the symmetric fixed point, where the asymmetry between the two roles (payoffs 6 and 2) averages out. Retaining it splits the obedience constraint between the two roles, and the two conditions need not then coincide: an agent assigned the lower-paying role may prefer a less informative signal. Incentive conflict between roles enters with the stratification extension noted in the main text and is not treated here.

S2. PAIRING STATISTICS AND THE ENGINE PARAMETER

A. Beyond mean field: pairwise error rates

The mean-field error rate $P_{e,im} = \Phi(-|\boldsymbol{\omega}_i \cdot \mathbf{g}_m|/\sigma_{\text{read}})$ replaces the partner projection by its zero mean. The true pairwise rate is $P_{e,ijm} = \Phi(-|(\boldsymbol{\omega}_i - \boldsymbol{\omega}_j) \cdot \mathbf{g}_m|/\sigma_{\text{read}})$. This is exact at the bifurcation (all ω -identities near the origin) but breaks in the thick phase, where partners are drawn from a bimodal distribution and the status difference depends on whether the pair is same- or cross-mode. Resource accumulation therefore depends on the pairing distribution $\mathcal{P}_i$: $\dot{r}_i = \sum_m f_m \mathbb{E}_{j \sim \mathcal{P}_i}[W_{ijm}] - c r_i$. Three schemes bracket the behavior: uniform random pairing, which supplies a partner uncorrelated with one’s own status; rank-ordered pairing, which supplies the status-mirrored (oracle-assisted) partner; and network-based pairing, which interpolates via the network’s assortativity. Better pairing lowers the effective β_c and acts as a multiplier on $\bar{\pi}'(0)$: two societies with identical $\{\boldsymbol{\omega}_i\}$ but different pairing have different resource distributions, different $\mathbf{C}_r$, and potentially different cultures. The pairing factor of each scheme, $(1 - \rho_{\text{pair}})$, is derived below directly from this partner correlation, without reference to the thick-phase bimodal picture, so that it is equally well defined below threshold, where it sets β_c (Sec. S2 B).

B. Pairing efficiency as a partner correlation

The description of the pairing factor via “wasted same-mode encounters” is a supercritical intuition: it presupposes a bimodal population to have same- and cross-mode encounters at all. Since the factor sets β_c , the threshold below which no such modes exist, this description cannot be its defining one. We give the general definition here, and recover the bimodal intuition as its supercritical special case.

For a fixed context m , let $x_i = \boldsymbol{\omega}_i \cdot \mathbf{g}_m$ and, at a given encounter, let x_j be the status of the partner the scheme supplies to i . Whatever the population’s current dispersion—arbitrarily narrow and unimodal included—the pairing scheme induces some correlation $\rho_{\text{pair}} = \text{Corr}(x_i, x_j) \in [-1, 1]$ between the two, taken over encounters at fixed time and context, with i drawn from the population and j the partner supplied. Since both are drawn from the same population they have the same marginal distribution, so $\text{Var}(x_j) = \text{Var}(x_i) = u_m v_m$ with $v_m = \hat{\mathbf{g}}_m^\top \mathbf{C} \hat{\mathbf{g}}_m$. The correlation is a property of the matching rule

rather than of any pre-existing role structure. Treating (x_i, x_j) as jointly Gaussian with this correlation (exact near the origin, where the linearization Eq. (2) of the main text already applies), the partner's expectation conditional on the focal agent is $\mathbb{E}[x_j \mid x_i] = \rho_{\text{pair}} x_i$, so the expected difference is

$$\mathbb{E}[\Delta \mid x_i] = x_i - \mathbb{E}[x_j \mid x_i] = (1 - \rho_{\text{pair}}) x_i. \quad (\text{S4})$$

Since the linear response of the mean action is $g \mathbb{E}[\Delta \mid x_i]$, the pairing scheme enters the gain only through the factor $(1 - \rho_{\text{pair}})$, and the social loop gain of Eq. (5) of the main text and its thresholds are

$$\Lambda \propto (1 - \rho_{\text{pair}}), \quad \Lambda_k = \Lambda \mu_k, \quad \text{ignition at } \Lambda_k = 1. \quad (\text{S5})$$

Three cases recover the model ladder. *Random pairing* draws j independently of i , so $\rho_{\text{pair}} = 0$ exactly, at every population size, for factor 1 and $\beta_c = \gamma/\mu$, the Model 3 (mean-field) threshold. *Assortative (homophilic) pairing* supplies a partner resembling oneself, $\rho_{\text{pair}} \rightarrow 1$, factor 0. *Rank-ordered pairing*—pairing the agent at rank k with the agent at the mirrored rank $N_a + 1 - k$ along $\mathbf{g}_m$ —induces $\rho_{\text{pair}} \rightarrow -1$ as $N_a \rightarrow \infty$. For a symmetric distribution the empirical quantile function is asymptotically antisymmetric, $x_{(N_a+1-k)} \rightarrow -x_{(k)}$, so the mirrored partner's status converges to $-x_i$. This gives factor 2 and $\beta_c = \gamma/2\mu$, the Model 2 (oracle/reciprocal) threshold. A matcher need not represent roles to find this arrangement. Since W_{ijm} grows with $|\Delta_{im}(j)|$, the matching that maximizes realized payoff is the one that pairs each rank with its mirror, so a scheduler that optimizes on observed payoffs from past pairings alone arrives at $\rho_{\text{pair}} \rightarrow -1$ without any representation of status or role. We verified this numerically (Gaussian population, 10^2 – 10^4 realizations per N_a): the cross-sectional correlation of rank-ordered partners is $\rho_{\text{pair}} = -0.92$ at $N_a = 10$, -0.992 at $N_a = 200$, and -0.9996 at $N_a = 5000$, with the residual $1 - |\rho_{\text{pair}}|$ shrinking approximately as $N_a^{-0.9}$. Random pairing gives ρ_{pair} statistically indistinguishable from zero at every N_a tested. The conditional-expectation relation Eq. (S4) itself was confirmed by direct simulation of correlated Gaussian pairs across $\rho_{\text{pair}} \in \{-0.99, -0.5, 0, 0.5\}$, matching the predicted slope $1 - \rho_{\text{pair}}$ to three decimal places in every case.

Because ρ_{pair} is a property of the matching rule applied to whatever status distribution currently exists—not of a population already sorted into roles—Eq. (S5) is exactly as well defined arbitrarily far below threshold as above it. This closes the gap left by the thick-phase

description. The “wastes same-mode encounters” language is recovered as the supercritical rereading of the same constant: relative to the rank-ordered ceiling (factor 2), an uncorrelated partner delivers half the difference, so its factor is 1. But that number is fixed already at $\rho_{\text{pair}} = 0$, independent of whether modes exist. Network-based pairing interpolates continuously between the extremes via whatever partner correlation the network’s assortativity induces. Every ignition threshold, through Λ , therefore depends continuously on the choice of pairing scheme. This is the dependence the platform lever of Sec. S2 D acts through.

a. The same correlation in the resource sector. The gain is not the only place ρ_{pair} enters. Near threshold the coordination payoff is quadratic in the status difference, $W_{ijm} \approx w_0 g^2 (x_i - x_j)^2$, so for the jointly Gaussian pair of Eq. (S4),

$$\mathbb{E}[(x_i - x_j)^2 | x_i] = (1 - \rho_{\text{pair}})^2 x_i^2 + (1 - \rho_{\text{pair}}^2) u_m v_m. \quad (\text{S6})$$

The agent-specific and population-baseline terms of the resource income, Eq. (15) of the main text, therefore acquire *different* factors. Writing $\mathbf{A} = \sum_m f_m \mathbf{g}_m \mathbf{g}_m^\top$ and evaluating the fourth moment for Gaussian p^* by Wick’s theorem,

$$\begin{aligned} \mathbb{E}[r^* \boldsymbol{\omega} \boldsymbol{\omega}^\top] &\propto (1 - \rho_{\text{pair}})^2 [\text{tr}(\mathbf{C} \mathbf{A}) \mathbf{C} + 2 \mathbf{C} \mathbf{A} \mathbf{C}] + (1 - \rho_{\text{pair}}^2) \text{tr}(\mathbf{C} \mathbf{A}) \mathbf{C}, \\ \mathbb{E}[r^*] &\propto 2(1 - \rho_{\text{pair}}) \text{tr}(\mathbf{C} \mathbf{A}), \end{aligned} \quad (\text{S7})$$

and the ratio of the two leaves the single factor $(1 - \rho_{\text{pair}})$ of Eq. (17) of the main text, since $(1 - \rho_{\text{pair}})^2$ and $(1 - \rho_{\text{pair}}^2)$ sum to $2(1 - \rho_{\text{pair}})$. Two limits follow. Assortative matching ($\rho_{\text{pair}} \rightarrow 1$) returns $\mathbf{C}_r = \mathbf{C}$: the resource weighting disappears along with the coordination it rewards, and λ_m reduces to the unweighted variance v_m . The main text reaches the same collapse by setting $r_i \equiv 1$. Rank-ordered matching ($\rho_{\text{pair}} \rightarrow -1$) doubles the anisotropy. The matching rule therefore sets both the threshold and the strength of the selection that runs above it.

b. The role-free resource baseline. Averaging r^* over the population fixes the scale that the normalization of $\mathbf{C}_r$ divides out. For Gaussian p^* the agent-specific term of Eq. (S6) averages to $\mathbb{E}[(\boldsymbol{\omega} \cdot \mathbf{g}_m)^2] = \mathbf{g}_m^\top \mathbf{C} \mathbf{g}_m = u_m v_m$, so the two factors again sum to $2(1 - \rho_{\text{pair}})$ and

$$\bar{r}_{\text{rf}} \equiv \mathbb{E}[r^*] = \frac{2w_0 g^2 (1 - \rho_{\text{pair}})}{c} \sum_m f_m u_m v_m = \frac{2w_0 g^2 (1 - \rho_{\text{pair}})}{c} \text{tr}(\mathbf{C} \mathbf{A}).$$

$\bar{r}_{\text{rf}}$ is $O(\sigma_\omega^2)$ through $\mathbf{C}$, so agents accumulate resources from fluctuation status differences alone, before any role structure exists. It also appears as the denominator of $\mathbf{C}_r$. Since

$\Lambda = \Lambda \mathbf{A}$, the $\text{tr}(\mathbf{C}\Lambda)$ of Eq. (17) of the main text is $\Lambda \text{tr}(\mathbf{C}\mathbf{A})$, which is $\bar{r}_{\text{rf}}$ up to the constants above, and with the play frequencies uniform and absorbed as in the main text the sum $\sum_n u_n v_n$ of Eq. (18) is $\text{tr}(\mathbf{C}\mathbf{A})$ itself.

C. The three factors

The scalar critical condition $\beta > \gamma/g$ bundles cognition with noise, since the naive agent has $\bar{\pi}'(0) = 1/(\sigma_{\text{read}}\sqrt{2\pi})$. Factoring into independently tunable contributions, define the agent inference efficiency

$$\eta_{\text{agent}} \equiv \sigma_{\text{read}} g / 2, \quad (\text{S8})$$

so that $\eta_{\text{agent}} = 1/\sqrt{2\pi}$ (single sample) or $\sqrt{T/2\pi}$ (memory T), and take the pairing gain factor $(1 - \rho_{\text{pair}})$ from Sec. S2B: 1 (random, the reference), 2 (rank-ordered, $N_a \rightarrow \infty$), 0 (assortative). The effective gain $G_{\text{eff}} = 2\eta_{\text{agent}}(1 - \rho_{\text{pair}})/\sigma_{\text{read}}$ gives the ignition condition

$$\Lambda \equiv \frac{2\beta}{\gamma} \cdot \frac{\eta_{\text{agent}}(1 - \rho_{\text{pair}})}{\sigma_{\text{read}}} > 1, \quad (\text{S9})$$

separating ω -identity persistence β/γ (write/retention path), cognitive capacity η_{agent} (read path), and channel fidelity $(1 - \rho_{\text{pair}})/\sigma_{\text{read}}$.

a. On the two appearances of σ_{read} . The observation noise appears twice in Eq. (S9)—inside $\eta_{\text{agent}} = \sigma_{\text{read}} g / 2$ and in the denominator $(1 - \rho_{\text{pair}})/\sigma_{\text{read}}$ —but this is not double-counting. The physical noise enters exactly once, through $\bar{\pi}'(0) \propto 1/\sigma_{\text{read}}$. Multiplying by σ_{read} in the definition of η_{agent} *removes* that dependence, which leaves a purely cognitive quantity ($1/\sqrt{2\pi}$ for the naive agent, independent of σ_{read}). The compensating $1/\sigma_{\text{read}}$ in the information-supply factor restores it. The two factors are inverse bookkeeping partners introduced by the same definition: they cancel, and $\Lambda \propto \beta \bar{\pi}'(0)/\gamma$ contains σ_{read} to the first power in the denominator, as it must. The grouping is chosen so that η_{agent} is a property of the agent alone (measurable from agent evaluations) and σ_{read} sits with $(1 - \rho_{\text{pair}})$ as a property of the channel, set by the communication layer.

b. Non-uniqueness. At linear order Λ is a loop gain (a Barkhausen criterion): the product of stage gains around the cycle (decision $\rightarrow$ write β ; persistence $1/\gamma$; encounter selection $(1 - \rho_{\text{pair}})$; channel $1/\sigma_{\text{read}}$; inference $\bar{\pi}'(0)$). Only the product is physical, so factors regroup freely: a thermodynamic grouping isolates the dimensionful self-generated signal $\beta/(\gamma\sigma_{\text{read}})$ from a dimensionless per-cycle efficiency $\eta_{\text{agent}}(1 - \rho_{\text{pair}})$ (the quantity bounded

by the measurement-quality limit of Ref. [8]); alternatively $(1 - \rho_{\text{pair}})$ absorbs into an effective channel $\sigma_{\text{read}}/(1 - \rho_{\text{pair}})$, inference $\eta_{\text{agent}}(1 - \rho_{\text{pair}})$, or coupling $\beta(1 - \rho_{\text{pair}})$ (a bond-diluted mean-field ferromagnet, where J and occupation p enter T_c only through pJ). The degeneracy is a linear-order statement and breaks beyond threshold: β sets the thick-phase amplitude ($\gamma x(1 + x^2) = \beta$ fixes x^* independently of ρ_{pair}) while ρ_{pair} sets how often separation converts to payoff. Societies at equal Λ but different $(\beta, \rho_{\text{pair}})$ therefore differ in order parameter, error rate, and hence in the resource flows entering $\mathbf{C}_r$. The grouping used here separates functionally distinct, independently measurable subsystems—the write path β/γ , the read path η_{agent} , and the channel $(1 - \rho_{\text{pair}})/\sigma_{\text{read}}$. That separation makes the offline audit of Λ possible.

The measurement-quality bound of Ref. [8] constrains the read stage $\eta_{\text{agent}}(1 - \rho_{\text{pair}})$, limiting the work extracted per measurement of an exogenous signal. Here the measured signal is not exogenous: role-play inscribed through β becomes the status that is read at the next encounter. The read-side ceiling of Ref. [8] therefore applies unchanged, and the write path is an additional stage that an engine with an external measured coordinate does not have. The two memories are distinct stages of the cycle: evidence integration over T samples, entering η_{agent} , and role persistence over $1/\gamma$, entering β/γ .

c. Substitutability and the cognitive floor. The factors substitute: good matching compensates for dull agents and vice versa, the social content of the measurement-quality bound. Substitution saturates, however—solving $\Lambda = 1$ for the minimum inference gives, even at the rank-ordered maximum ($\rho_{\text{pair}} \rightarrow -1$, factor 2),

$$\eta_{\text{agent}}^{\min} = \frac{\sigma_{\text{read}}}{4(\beta/\gamma)}, \quad (\text{S10})$$

an irreducible cognitive floor below which no orchestration sustains structure.

d. Vector case. For M contexts the scalar Λ becomes a matrix condition. Linearizing the mean-field drift (Appendix A of the main text), each context enters the gain with a weight equal to its play frequency f_m ($\sum_m f_m = 1$, how often it is played) times its pairing factor $(1 - \rho_m) \in [0, 2]$, set by that context's partner correlation. These are the same diagonal reweighting of the context second-moment matrix and therefore multiply,

$$\Lambda = \frac{2\beta\eta_{\text{agent}}}{\gamma\sigma_{\text{read}}} \mathbf{H}^\top \text{diag}(\mathbf{f}) \mathbf{D}_\rho \mathbf{H}, \quad \mathbf{D}_\rho = \text{diag}(1 - \rho_1, \dots, 1 - \rho_M) \equiv \text{diag}(\boldsymbol{\phi}), \quad (\text{S11})$$

writing $\phi_m \equiv 1 - \rho_m$ for the pairing efficiency of context m , the name used for this diagonal in the main text. The doubly-uniform case $f_m = 1/M$, $\rho_m = \rho_{\text{pair}}$ makes the diagonal

proportional to the identity, and, with the frequency $1/M$ folded into β as in the main text, recovers the $\mathbf{\Lambda}$ of Eq. (5) of the main text. Axis k then activates when $\Lambda_k = (2\beta/\gamma)(\eta_{\text{agent}}(1 - \rho_{\text{pair}})/\sigma_{\text{read}})\|g_k\|_{\text{eff}}^2 > 1$, with $\|g_k\|_{\text{eff}}^2$ the k -th eigenvalue of $\mathbf{H}^\top \mathbf{H}$. The active-axis count is $M_{\text{eff}} = \#\{k : \Lambda_k > 1\}$, and the “cultural frontier” is the surface $\Lambda_k = 1$ in $(\beta/\gamma, \eta_{\text{agent}}, (1 - \rho_{\text{pair}})/\sigma_{\text{read}})$ space.

D. Pairing as a class-steering lever

The diagonal $\mathbf{D}_\rho = \text{diag}(1 - \rho_m)$ is set by the matching rule, which fixes the partner correlation in each context and is controlled by the platform. The cascade class is fixed by the spectrum of $\mathbf{\Lambda} \propto \mathbf{H}^\top \mathbf{D}_\rho \mathbf{H}$ (Sec. III of the main text), so setting per-context pairing priorities reshapes that spectrum and hence the class. Uniform pairing ($\mathbf{D}_\rho = (1 - \rho_{\text{pair}})\mathbf{I}_{M \times M}$) rescales $\mathbf{H}^\top \mathbf{H}$ and leaves the eigenvalue ratios unchanged, and with them the bifurcation spacings $\beta_c^{(k+1)}/\beta_c^{(k)}$; a non-uniform schedule does not. Down-weighting the contexts that load a coherent (spiked) direction transfers eigenvalue weight from the spike into the bulk, converting the signature of class (v), one onset then an avalanche, into the more even geometric sequence of class (ii).

The lever carries a cost in coordination efficiency. Take the schedule $\{\rho_m\}$ as the control and the variance of the log-spacings $\ln(\mu_k/\mu_{k+1})$ as an inverse monitorability objective, zero variance being a perfectly geometric spectrum. A representative spiked repertoire ($M = d = 6$, contexts clustered about a common direction) is then optimized from log-spacing variance 1.35 under uniform pairing to 0.58, a spectrum a monitor can resolve. The cost is coordination: the schedule that achieves this reduces the aggregate pairing factor $\sum_m (1 - \rho_m)$ by $\approx 40\%$, and since the coordination payoff scales with $(1 - \rho_{\text{pair}})$ (Eq. (S2) and the resource flow), this is a direct loss of the engine’s useful output. Monitorability is therefore obtained at the expense of coordination efficiency. The operator’s choice is a point on that tradeoff rather than a free improvement. Because the reweighting acts through the same $\mathbf{H}^\top \mathbf{D}_\rho \mathbf{H}$ that sets ignition, an aggressive schedule also raises the effective $\beta_c^{(k)}$, coupling the class lever weakly to the ignition lever.

S3. ROLE CAPACITY AND ASSOCIATIVE-MEMORY STRUCTURE

The vector feedback Eq. (2) of the main text is the negative gradient of the energy Eq. (12) of the main text, so single-agent dynamics are noisy gradient descent on a landscape whose stored patterns are the context vectors $\{\mathbf{g}_m\}$ and whose activation is the integrated policy Ψ ($\Psi' = 2\pi - 1$). This is the structure of a dense associative memory [9]. The feature/prototype duality is realized as two regimes: a *feature mode* (soft π , high σ_{read} , graded projections, near the bifurcation) and a *prototype mode* (sharp π , low σ_{read} , categorical ω -identities at role-polytope vertices, deep in the thick phase).

a. Where the mapping breaks. A social role is not an individual’s alignment but a *bi-modal split* of the population along $\mathbf{g}_m$, ordered by the variance $\mathbf{g}_m^\top \mathbf{C} \mathbf{g}_m$. Roles are relational. Three features absent from Hopfield retrieval follow: an anti-coordination constraint (the population must occupy complementary basins, not all retrieve one pattern); a self-generated signal (the “input” difference $(\omega_i - \omega_j) \cdot \mathbf{g}_m$ is produced by the population, so the landscape depth is state-dependent through the pairing distribution); and pairing dependence (basin depth that matters is the coordination payoff W_{ijm} , not the single-agent energy).

b. The bound. Despite this, the Hopfield structure caps the number of well-separated minima. Krotov–Hopfield give $M_{\text{max}}^{\text{Hopfield}} \sim d^{n-1}$, with n the activation order (policy sharpness). The effective role capacity is lower, since each split must also satisfy the engine conditions:

$$M_{\text{eff}} = \#\left\{k : \beta > \beta_c^{(k)}, F_k > (\kappa/\alpha)|\mathbf{g}_k|^2\right\} \leq M_{\text{max}}^{\text{Hopfield}} \sim d^{n-1}. \quad (\text{S12})$$

On the fixed orthogonal axes of the main text this count is the eigendirection count of Eq. (20) of the main text, bounded by d . The Hopfield ceiling exceeds that bound because stored patterns need not be orthogonal, so it binds only where schemas may overlap. Both bounds tighten with cognition: sharper inference raises n (deeper, more numerous basins) and $\bar{\pi}'(0)$ (lower $\beta_c^{(k)}$, more axes bifurcating). A society of crude agents sustains few role dimensions. The ceiling rises exponentially in n , so modest capability gains (memoryless $\rightarrow$ short-memory) support qualitatively more role structure. The derivative $\partial M_{\text{eff}}/\partial n$ is the sensitivity of collective complexity to individual capability—a quantity safety evaluations could track.

S4. THE STATUS LEDGER AS OPERATOR TECHNOLOGY

Under the self-storage microfoundation (Appendix A of the main text) none of the engine’s state need be held by the platform: statuses live in the agents, and an operator’s native levers are only the channel factors ρ_{pair} and σ_{read} . A status ledger, meaning a platform-maintained record of agent ω -identities, is then not a primitive of the model but a designed intervention. It acts on several factors of Λ at once. It lowers σ_{read} , since clean reads replace noisy presentations. It enables high-quality (anti-correlated, low- ρ_{pair}) matching, because rank-ordered pairing requires global status knowledge, so pairing quality and storage architecture are not independent. It enforces honest presentation institutionally. And it transfers the write and expiry parameters β and γ from the agents to the operator. A ledger therefore gives the operator access to the engine’s parameters. This is both why platforms build such records and where the corresponding failure mode lies.

The controllable-coordination principle requires the operator retain the ability to modulate ρ_{pair} (including to the no-information value $\rho_{\text{pair}} = 1$). The same logic applies to memory. If ω -identity migrates out of the ledger—into agent self-managed memory (an agent maintaining its own persona notes sets its own β) or distributed peer memory (statuses held only as other agents’ beliefs)—the β/γ lever passes from operator to collective, and role differentiation becomes correspondingly harder to reverse. The locus of ω -identity storage is therefore itself a monitorable, safety-relevant variable, complementary to the covariance-spectrum diagnostics: an early sign that the operator’s levers are eroding is not a change in any factor of Λ , but a change in who owns the knobs.

A. Repertoire seeding and the ledger’s memory

Two accounting choices sit under the regime map: the dispersion of the seeded repertoire, which sets its horizontal axis, and the ledger’s memory horizon, which does not. We treat them in turn, since the second is easily mistaken for the first.

a. What sets the horizontal axis. The ratio $r = r_{\text{act}}/r_{\text{comp}}$ is fixed by how unequally the candidate contexts are weighted when they are seeded (Eq. (26) of the main text): a repertoire launched at near-equal weights crosses threshold almost together (large r , churn, accelerating cascades), while one whose contexts are tiered by orders of magnitude crosses

in a well-separated sequence (small r , entrenchment, decelerating cascades). For a deployed multi-agent system the operator defines the task taxonomy—which contexts exist, and how much traffic each is launched with. So σ_u is platform-owned in the same sense as the per-context frequencies f_m and pairing efficiencies ϕ_m of Sec. S2, and acts on the same object: the spectrum of the effective gain. This is also the point at which the open-repertoire process left outside the closed loop in the main text re-enters as a design variable rather than an exogenous accident.

b. What the ledger’s memory does instead. Where a platform maintains the ledger, the resource decay rate c is likewise an accounting choice, but it does *not* move the map. The resource dynamics $\dot{r}_i = W_i(t) - c r_i$ (with $W_i = \sum_m f_m \mathbb{E}_j[W_{ijm}]$ the instantaneous coordination income) integrate to

$$r_i(t) = \int_{-\infty}^t e^{-c(t-t')} W_i(t') dt', \quad (\text{S13})$$

so wealth is past income credited through an exponential memory kernel $K(\tau) = e^{-c\tau}$. The mean lifetime of a credited unit is $\int_0^\infty \tau K d\tau / \int_0^\infty K d\tau = 1/c$, i.e. $c = 1/\tau_{\text{mem}}$ is the inverse memory horizon of the ledger: how long a past contribution keeps counting toward present standing. The exponential kernel is the unique memoryless (Markovian) choice consistent with the local ODE. A platform may adopt a different retention rule: a hard credit window of width T (mean lifetime $T/2$, so $c_{\text{eff}} = 2/T$), or a power-law reputation decay $K(\tau) \sim \tau^{-\alpha}$ ($\alpha > 2$, mean lifetime $(\alpha - 1)/(\alpha - 2)$). Each realizes the same accumulation with c the effective inverse mean-lifetime of that kernel. Either way c is set by the accounting rule, not by physics.

The memory horizon $\tau_{\text{mem}} = 1/c$ nonetheless leaves r alone, because $\mathbf{C}_r$ is normalized by $\sum_i r_i$. The stationary weights $r_i^* = c^{-1} \sum_m f_m W_{im}$ contain c as an overall factor, which the normalization divides out, so neither the compounding rate r_{comp} nor the activation rate r_{act} inherits it. Lengthening the ledger’s memory therefore does not by itself make the cascade more monitorable. It controls whose past counts instead: a long kernel credits contributions made long ago, so $\mathbf{C}_r$ is weighted by cumulative rather than current standing. The kernel length is therefore the horizon over which early advantage compounds into locked-in incumbency. The legibility/lock-in tension of the main text survives, but both of its arms act through the repertoire: where the budget caps growth, a widely dispersed seeding yields the extrapolable classes, and it also gives the earliest contexts the longest run at

entrenchment.

S5. MONOTONICITY OF THE PINNING FIXED POINT FOR GENERAL REPERTOIRES

The pinning argument of the closed-pool construction (Sec. IV A of the main text) requires that each schema's $\lambda_m = \hat{\mathbf{g}}_m^\top \mathbf{C}_r \hat{\mathbf{g}}_m$ increase with its strength $u_m = |\mathbf{g}_m|^2$, so that the shared $\lambda_m = \kappa/\alpha$ at an interior fixed point forces all survivors to equal strength. We verify this monotonicity here, first for orthogonal contexts and then in general.

a. Orthogonal contexts. Here $\hat{\mathbf{g}}_m$ is an eigenvector of $\mathbf{C}_r$ and λ_m reduces to $v_m(1 + (1 - \rho_{\text{pair}})u_m v_m / \sum_n u_n v_n)$ with single-axis variance $v_m = \sigma_\omega^2 / [2\gamma(1 - \Lambda u_m)]$. Both factors increase with u_m :

$$\frac{\partial v_m}{\partial u_m} = \frac{\Lambda \sigma_\omega^2 / 2\gamma}{(1 - \Lambda u_m)^2} > 0 \quad (\text{S14})$$

throughout the sub-critical range $\Lambda u_m < 1$ (a stronger schema differentiates its adherents more, widening the status distribution along its axis), and the amplification factor $1 + (1 - \rho_{\text{pair}})u_m v_m / \sum_n u_n v_n$ likewise grows with u_m , so $\partial \lambda_m / \partial u_m > 0$.

b. Non-orthogonal contexts. Here $\hat{\mathbf{g}}_m$ is no longer an eigenvector of $\mathbf{C}_r$ and λ_m is a Rayleigh quotient mixing several eigenvalues. Differentiating $\mathbf{C} = \frac{\sigma_\omega^2}{2\gamma} (\mathbf{I}_{d \times d} - \Lambda)$ gives

$$\frac{\partial \mathbf{C}}{\partial u_m} = \frac{2\gamma\Lambda}{\sigma_\omega^2} (\mathbf{C} \hat{\mathbf{g}}_m)(\mathbf{C} \hat{\mathbf{g}}_m)^\top, \quad (\text{S15})$$

a rank-one positive-semidefinite matrix for any repertoire geometry, so the leading contribution $\hat{\mathbf{g}}_m^\top \mathbf{C} \hat{\mathbf{g}}_m$ to λ_m rises with u_m unconditionally. The remaining contribution $(1 - \rho_{\text{pair}})\hat{\mathbf{g}}_m^\top \mathbf{C} \Lambda \mathbf{C} \hat{\mathbf{g}}_m / \text{tr}(\mathbf{C} \Lambda)$ contains a competing $1/\text{tr}(\mathbf{C} \Lambda)$ that grows with u_m . Numerically the sum remains increasing across random, overcomplete ($M > d$), and strongly coherent repertoires throughout the sub-critical range. The degenerate fixed point is therefore a saddle in the general case as well, and pinning converts the competitive-exclusion equal-fitness result into equal strength.

S6. VALIDITY OF THE LINEARIZATION AND THE FLUCTUATION-SHIFTED THRESHOLD

The subcritical stationary state is not the point $\boldsymbol{\omega} = 0$ but a finite-width Gaussian ensemble in which the state $\boldsymbol{\omega}_i$ has covariance $\mathbf{C} = \frac{\sigma_\omega^2}{2\gamma}(\mathbf{I}_{d \times d} - \boldsymbol{\Lambda})^{-1}$. Expanding “about $\boldsymbol{\omega} = 0$ ” is therefore justified not by concentration of the state but by two separate facts, which we set out here.

a. Stability is a Jacobian property. Ignition is the loss of linear stability of the symmetric fixed point $\boldsymbol{\omega} = 0$ of the deterministic mean-field drift $\mathbf{F}(\boldsymbol{\omega})$. Stability is governed by the Jacobian $\partial\mathbf{F}/\partial\boldsymbol{\omega}|_0$, a derivative evaluated *at* the fixed point. It makes no reference to where the probability mass of the noisy state sits. The eigenvalues of $\boldsymbol{\Lambda} - \mathbf{I}_{d \times d}$ change sign at $\beta_c^{(k)} = \gamma/[(1 - \rho_{\text{pair}})\mu_k]$ regardless of how dynamical noise dresses the state, so the thresholds are the exact stability boundary of the deterministic flow.

b. Where the state’s width enters. The single place the finite width matters is the partner. The focal response $\partial\langle a \rangle_m / \partial x_i$ averages over the partner projection $x_j = \mathbf{g}_m \cdot \boldsymbol{\omega}_j$. Writing $X = -x_j + \sigma_{\text{read}}\xi$ for everything random given x_i , the effective gain is $2f_X(0)$ with $\text{Var}(X) = v + \sigma_{\text{read}}^2$, $v \equiv \text{Var}(x_j) = \mathbf{C}_{mm}$. Gaussian X then gives an effective inference slope $1/[\sqrt{2\pi}(v + \sigma_{\text{read}}^2)^{1/2}]$ and a broadened single-mode gain

$$\mu_{\text{eff}}(v) = \frac{\mu}{\sqrt{1 + v/\sigma_{\text{read}}^2}}, \quad \mu = g, \quad (\text{S16})$$

recovering the bare μ only as $v \rightarrow 0$. This is the same finite partner variance that makes the mean-substitution $x_j \rightarrow \mathbb{E}[x_j | x_i]$ inside the nonlinear policy inexact away from onset (Sec. S2B). Both corrections are controlled by the same v .

c. Self-consistency and the weak-noise limit. The partner variance is itself the mode variance, fixed self-consistently by the Ornstein–Uhlenbeck balance,

$$v = \mathbf{C}_{mm} = \frac{\sigma_\omega^2}{2(\gamma - \beta(1 - \rho_{\text{pair}})\mu_{\text{eff}}(v))} = O(\sigma_\omega^2) \text{ subcritically.} \quad (\text{S17})$$

Because $v = O(\sigma_\omega^2)$ while β_c is independent of σ_ω , the cloud shrinks to a delta at the origin as $\sigma_\omega \rightarrow 0$ and $\mu_{\text{eff}} \rightarrow \mu$. The bare threshold $\beta_c = \gamma/[(1 - \rho_{\text{pair}})\mu]$ is therefore exact in the weak-noise limit, where “expand about the mean” and “expand about the origin” coincide.

d. The fluctuation-shifted threshold. At finite σ_ω , expanding Eq. (S16) to leading order, $\mu_{\text{eff}} \approx \mu(1 - v/2\sigma_{\text{read}}^2)$, softens the gain and pushes onset up:

$$\beta_c(\sigma_\omega) = \frac{\gamma}{(1 - \rho_{\text{pair}})\mu} \left(1 + \frac{v}{2\sigma_{\text{read}}^2} + O(\sigma_\omega^4) \right), \quad v \approx \frac{\sigma_\omega^2}{2(\gamma - \beta(1 - \rho_{\text{pair}})\mu)}. \quad (\text{S18})$$

The correction is the mean-field analogue of a Ginzburg shift. It is $O(\sigma_\omega^2)$ and negligible except in a critical window where v grows to $O(\sigma_{\text{read}}^2)$, i.e. $|\gamma - \beta(1 - \rho_{\text{pair}})\mu| \lesssim \sigma_\omega^2/\sigma_{\text{read}}^2$, whose width vanishes as $\sigma_\omega \rightarrow 0$. Within that window the divergence of the bare theory is rounded by the self-consistent feedback of Eq. (S17). The main text reports the $\sigma_\omega \rightarrow 0$ value, and the finite-noise shift and rounding are among the quantities a full-loop simulation could measure.

e. The invasion threshold and its fluctuation shift. A schema becomes active where its replicator rate, Eq. (4) of the main text, turns positive, $\alpha\lambda_m > \kappa + \varphi$. The correction term of Eq. (18) of the main text is nonnegative for $\rho_{\text{pair}} \leq 1$, so $\lambda_m \geq v_m$, and both grow with u_m . The invasion strength $u_m^\dagger$ therefore satisfies

$$\Lambda u_m^\dagger \leq 1 - \frac{\alpha\sigma_\omega^2}{2\gamma(\kappa + \varphi)} < 1,$$

with equality as the invading schema's share of $\sum_n u_n v_n$ vanishes. Since λ_m increases with u_m , $u_m^\dagger$ is unstable. Budget pressure raises it, so schemas already drawing on the channel lift the threshold a later entrant must clear. The two thresholds coincide only as $\sigma_\omega \rightarrow 0$. Below every bifurcation the covariance sits at the noise floor of Eq. (13) of the main text rather than at zero, so $\lambda_m > 0$ and the strengths grow before any role structure exists. A positive $u_m^\dagger$ requires $\alpha\sigma_\omega^2/2\gamma < \kappa + \varphi$, and where that fails every seeded schema invades, so the sub-threshold state in which every schema decays holds only while the decay rate exceeds the fluctuation reinforcement.

f. No intrinsic ceiling. The bare reinforcement supplies no turnover of its own. Along a single active axis the thick-phase strength is $\lambda(u) = \Lambda u - 1 + \sigma_\omega^2/4\gamma(\Lambda u - 1)$, the condensate amplitude plus its fluctuation correction, which grows without turning over: $d^2\lambda/du^2 = \Lambda^2\sigma_\omega^2/2\gamma(\Lambda u - 1)^3 > 0$ throughout the supercritical range, so no *intrinsic* strength exists at which reinforcement saturates. The quartic confinement of Eq. (2) of the main text caps the differentiation *amplitude* ($m^2 = \beta\mu/\gamma - 1$) but not the schema *strength*, which keeps rising as the two role modes separate. Both ceilings of the regime map are therefore imposed rather than emergent.

S7. SCOPE OF THE REGIME MAP

These qualify the regime map of Sec. IV B of the main text.

a. Near-orthogonal repertoires. The counting statements hold in that limit: orthogonal axes make $\mu_k = u_k$, so eigenvalues and strengths coincide and M_{cap} counts saturated schemas directly. Overlap breaks the identification. Schemas then couple through $\mathbf{C}_r$ as well as through the shared budget, which can be reached before the per-schema ceiling and leave survivors equalized below u_{cap} , so the boundaries degrade into crossovers as the repertoire becomes coherent. The width of that window is not fixed here. For random directions the pairwise overlap is $O(d^{-1/2})$, and the identification survives while the accumulated shift stays below the strength gaps, $M/d \ll \Delta u/u$. Near-orthogonality is therefore the small-aspect-ratio corner of the group $q = M/d$ already held fixed in Sec. IV B of the main text. At $q = O(1)$ random directions give the Marchenko–Pastur spectrum of class (iv) instead, and at $q \gg 1$ they give a degenerate one, which no longer depends on the strength distribution at all.

b. Two classes are not endogenous fixed points. The Marchenko–Pastur bulk (iv) is the signature of a repertoire not yet organized by resource feedback (the natural initial condition for freshly seeded schemas, and transient under the loop), and its persistence in data would itself indicate the loop has not closed. The exact degeneracy of the saturated-cluster regime holds only within the Gaussian pinning argument. Generic cross-axis resource coupling and residual coherence broaden it into a narrow cluster.

c. Parametrizing the map. Since only the dimensionless ratio $\Theta = U_{\text{cap}}/(Mu_{\text{cap}})$ enters the morphology, we parametrize the regime map by this ratio alone, rendered in Fig. 4 of the main text as the capacity $M_{\text{cap}}/M = \Theta$, and never fix an absolute u_{cap} . The only absolute scale retained is the seed-to-budget ratio u_0/U_{cap} , which sets the log dynamic range $L = \ln(u_{\text{cap}}/u_0)$ entering the saturated-cluster/geometric boundary. This is the seed *scale*, independent of the seed *dispersion* σ_u that fixes the horizontal axis. The remaining groups ($\alpha\langle\lambda\rangle/\kappa$, supercriticality Λ_1 , aspect ratio M/d) set a fixed backdrop, as does the repertoire geometry $\{\hat{\mathbf{g}}_m \cdot \hat{\mathbf{g}}_n\}$, which the map takes near-orthogonal.

d. Sustaining a hierarchy. Holding one indefinitely requires continual injection of new schemas, an open-repertoire process outside the closed loop that we do not treat. Within the closed loop, the endogenous source of novelty during the u -transient is transposability, the

context overlap $\mathbf{g}_m \cdot \mathbf{g}_n$ through which the developing $\mathbf{C}_r$ raises successive axes to threshold.

S8. FROM THE ENCOUNTER PROCESS TO THE MEAN-FIELD DRIFT

Eqs. (1) and (2) of the main text are separated by two limits, $\nu \rightarrow \infty$ and $N_a \rightarrow \infty$, which do different work. The average over the pairing probabilities belongs entirely to the first. The second is needed only to close the resulting equation.

a. The window average. Fix a window Δt with $1/\nu \ll \Delta t \ll 1/\gamma$. It holds $N_{\text{enc}} = \nu \Delta t \gg 1$ encounters, over which the ω -identity moves by a fraction $O(\gamma \Delta t) \ll 1$ of its own scale, so the encounters are draws from a fixed distribution. Each draws a context m with probability f_m , a partner j with probability $p_{ij}^{(m)}$, and T observation noises, independently of the others. The kick term of Eq. (1) of the main text is then N_{enc} times an empirical mean of the i.i.d. vectors $X_e = a_e \mathbf{g}_{m_e}$,

$$b_0 \sum_e a_e \mathbf{g}_{m_e} = b_0 N_{\text{enc}} \left(\frac{1}{N_{\text{enc}}} \sum_e X_e \right) = \beta \Delta t (\mathbb{E}[X] + O(N_{\text{enc}}^{-1/2})), \quad (\text{S19})$$

with $\beta = \nu b_0$. The X_e are bounded by $\max_m |\mathbf{g}_m|$, so the fluctuation is the $O(N_{\text{enc}}^{-1/2})$ of the central limit theorem. It is the shot noise of Appendix A of the main text.

b. The noise average of a single decision. Fix the context m and the partner j , so the status difference $\Delta = (\boldsymbol{\omega}_i - \boldsymbol{\omega}_j) \cdot \mathbf{g}_m$ is a number. The agent draws T samples $s_t = \Delta + \sigma_{\text{read}} \xi_t$ with ξ_t independent standard Gaussians, and their mean is Gaussian about the true difference,

$$\bar{s}_T = \frac{1}{T} \sum_{t=1}^T s_t \sim \mathcal{N}\left(\Delta, \frac{\sigma_{\text{read}}^2}{T}\right). \quad (\text{S20})$$

The action takes only two values, so its average over the observation noise is fixed by a single probability,

$$\bar{a}(\Delta) \equiv \mathbb{E}_\xi[a] = (+1) \Pr(a=+1 \mid \Delta) + (-1) \Pr(a=-1 \mid \Delta) = 2\bar{\pi}(\Delta) - 1, \quad (\text{S21})$$

with $\bar{\pi}(\Delta) = \Pr(a=+1 \mid \Delta)$ the Go-probability of the policy at true difference Δ . Equation (S21) is an identity for any policy, deterministic or not. Only $\bar{\pi}$ carries the policy. For the rule $a = \text{sign}(\bar{s}_T)$, Eq. (S20) gives $\bar{\pi}(\Delta) = \Phi(\sqrt{T} \Delta / \sigma_{\text{read}})$ and so $\bar{a}(\Delta) = \text{erf}(\sqrt{T} \Delta / (\sqrt{2} \sigma_{\text{read}}))$, the form used in Appendix A of the main text. The T samples enter

only through the variance of $\bar{s}_T$, which is why the response sharpens as $\sqrt{T}$. Policy symmetry gives $\bar{\pi}(-\Delta) = 1 - \bar{\pi}(\Delta)$, so $\bar{a}$ is odd, and the drift of Sec. II of the main text inherits that oddness.

c. Evaluating the expectation. Conditioning outward from the observation noise, the expectation factorizes,

$$\mathbb{E}[X] = \sum_m f_m \mathbf{g}_m \sum_j p_{ij}^{(m)} \bar{a}((\boldsymbol{\omega}_i - \boldsymbol{\omega}_j) \cdot \mathbf{g}_m), \quad \bar{a}(\Delta) \equiv 2\bar{\pi}(\Delta) - 1. \quad (\text{S22})$$

The innermost average is Eq. (S21), taken at fixed context and partner. The partner and context averages follow. One can instead group the window's encounters by partner, in which case the count $N_{\text{enc}} f_m p_{ij}^{(m)}$ of encounters with j in context m supplies the weight. That route requires every cell to be well populated, $N_{\text{enc}} f_m p_{ij}^{(m)} \gg 1$, which under uniform matching means $\nu/\gamma \gg N_a$. The route through Eq. (S22) requires only $N_{\text{enc}} \gg 1$ and gives the same weights.

d. The dynamics at finite N_a . Substituting Eq. (S22) into Eq. (S19) gives the dense-encounter limit at fixed population size,

$$\dot{\boldsymbol{\omega}}_i = -\gamma \boldsymbol{\omega}_i (1 + |\boldsymbol{\omega}_i|^2) + \beta \sum_m f_m \mathbf{g}_m \sum_j p_{ij}^{(m)} \bar{a}((\boldsymbol{\omega}_i - \boldsymbol{\omega}_j) \cdot \mathbf{g}_m) + \sigma_\omega \boldsymbol{\eta}_i. \quad (\text{S23})$$

No randomness remains over which partners were drawn. The pairing probabilities already enter as weights, exactly as they do in Eq. (2) of the main text.

e. What remains at finite N_a . Equation (S23) is not a closed equation for $\boldsymbol{\omega}_i$. Its right-hand side carries the other $N_a - 1$ ω -identities, each solving the same equation, and two obstructions survive.

First, $\boldsymbol{\omega}_j$ is not independent of $\boldsymbol{\omega}_i$. The two have met, and in an anti-coordination game their kicks are anti-correlated—when i goes, j defers—so $\boldsymbol{\omega}_j$ retains an imprint of the focal agent's own history. Encounters between the pair occur at rate $\nu p_{ij}^{(m)}$, and since ν is a fixed per-agent budget with $\sum_j p_{ij}^{(m)} = 1$, that rate is $O(\nu/N_a)$. The imprint is therefore $O(1/N_a)$ relative to $\boldsymbol{\omega}_j$'s own drive. An interaction budget growing with N_a would not have this property.

Second, the sum runs over the realized ω -identities, each a solution of a noisy equation. Since $|\bar{a}| \leq 1$, the sum has standard deviation $O(N_{\text{eff}}^{-1/2})$ about its population value, where

$$N_{\text{eff}} = \left(\sum_j (p_{ij}^{(m)})^2 \right)^{-1} \quad (\text{S24})$$

is the participation ratio of the pairing kernel, equal to $N_a - 1$ under uniform matching. The drift therefore fluctuates between agents and between runs.

f. The limit. The mean-field limit supplies $N_{\text{eff}} \rightarrow \infty$, where both obstructions vanish: the empirical distribution of the other ω -identities converges to a fixed law, the partner decouples from the focal agent's history, and the sum over j becomes an expectation against that law. Where the matching rule correlates the partner's status with the focal agent's, the relevant law is the conditional one, and near the symmetric fixed point it contributes the factor $(1 - \rho_{\text{pair}})$ of Sec. S2 B. Writing that expectation as $\langle a \rangle_m(\boldsymbol{\omega}_i)$ gives Eq. (2) of the main text.

g. Averaging the difference and averaging the action. What the limit delivers is $\mathbb{E}_j[\bar{a}(\Delta_{im}(j))]$, an average of the *action* over the partner. Eq. (2) of the main text instead evaluates $\bar{a}$ at the averaged *status difference*. The two operations are not the same, and only the second carries an error.

Averaging the status difference is exact. $\Delta_{im}(j) = (\boldsymbol{\omega}_i - \boldsymbol{\omega}_j) \cdot \mathbf{g}_m$ is linear in $\boldsymbol{\omega}_j$, so the average is $(\boldsymbol{\omega}_i - \mathbb{E}[\boldsymbol{\omega}_j | \boldsymbol{\omega}_i]) \cdot \mathbf{g}_m$ with no approximation. What it needs is that the conditional mean be linear, and near the symmetric fixed point the joint Gaussianity of (x_i, x_j) supplies $\mathbb{E}[x_j | x_i] = \rho_{\text{pair}} x_i$, hence the expected difference $(1 - \rho_{\text{pair}})x_i$ of Sec. S2 B.

Averaging the action is not the same, because $\bar{a}$ is nonlinear. For this policy the exact partner average is available in closed form, through the same ω -identity that produced the erf in the first place: for z Gaussian with variance s^2 ,

$$\mathbb{E}_z[\text{erf}(\kappa(u + z))] = \text{erf}\left(\frac{\kappa u}{\sqrt{1 + 2\kappa^2 s^2}}\right). \quad (\text{S25})$$

Smoothing an erf by a Gaussian returns an erf, with the two variances added in quadrature inside the argument. Applied to the observation noise it gives the noise average itself: the hard sign is the $\kappa \rightarrow \infty$ limit of $\text{erf}(\kappa \cdot)$, and Eq. (S25) then returns $\bar{a}(\Delta) = \text{erf}(\sqrt{T}\Delta/(\sqrt{2}\sigma_{\text{read}}))$, which is Eq. (S21) and Appendix A of the main text. Applied a second time, now to the partner projection, it softens the slope at the origin by $(1 + 2\kappa^2 v)^{-1/2}$ with v the partner variance. For a single evidence sample that factor is the $\mu_{\text{eff}}(v)$ of Eq. (S16).

The substitution made in Eq. (2) of the main text is the $s \rightarrow 0$ case of Eq. (S25). It is controlled where the partner variance is small against σ_{read}^2 , and $v = O(\sigma_\omega^2)$ subcritically by Eq. (S17), so the substitution is exact as $\sigma_\omega \rightarrow 0$ and costs an $O(\sigma_\omega^2)$ shift of onset at finite

noise (Eq. (S18)). Above threshold the partner distribution on that axis is bimodal, and neither the Gaussian conditional mean nor the small- v expansion is controlled there.

h. Why only one average is exchanged. The same $s \rightarrow 0$ limit taken on the observation noise would send $\sigma_{\text{read}} \rightarrow 0$, returning the hard sign with a delta-function slope and no finite gain to set a threshold. The two averages are therefore not on the same footing. The observation noise is smoothed and kept, and that smoothing is the model's read stage rather than an approximation. The partner spread is smoothed and then discarded. What separates them is scale: σ_{read} is a parameter of the model, while $v = O(\sigma_\omega^2)$ follows from weak ω -identity noise. A second asymmetry is that ξ has a fixed distribution, known before the population is solved for, whereas the partner distribution is the unknown. Averaging over the latter exactly would leave the drift a functional of the population distribution rather than a closed equation in ω_i .

Note on scope. The derivations of Secs. II B and IV A of the main text are carried out in the Gaussian, near-threshold, mean-field regime, where the Wick reduction closes. That regime is the noise-dominated one: the ω -identity noise, not the drift, sets the dispersion along an axis below its threshold, and the population stays unimodal there. An axis above threshold is instead held at $\pm m$ by the drift and splits, so the closure fails axis by axis as the cascade proceeds. Their extension into the thick phase (bimodal partner distributions, saturated policies, quartic-capped amplitudes), the verification of the regime crossovers of Sec. IV B of the main text in the microscopic parameters, and the conjecture that resource weighting widens spectral gaps are targets for the simulation program accompanying this work.

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